

SYDNEY TECHNICAL HIGH SCHOOL



EXTENSION 1

MATHEMATICS

PRELIMINARY ASSESSMENT TASK 2

JULY 2012

Time Allowed: 70 minutes

Instructions:

- Write using blue or black pen.
- Approved calculators may be used.
- Attempt all questions.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- **PLEASE START EACH NEW QUESTION ON A NEW PAGE.**

Name: _____

Teacher: _____

Q1	Q2	Q3	Q4	Q5	Q6	TOTAL
/10	/10	/10	/10	/10	/10	/60

Question 1

(10 marks)

- a) What is the value of $\tan\left(\frac{-3\pi}{4}\right)$? 1
- b) Find the exact value of $\cos 105^\circ$ 2
- c) Find the coordinates of the point which divides the interval AB,
with A(1, 4) and B(5, 2), externally in the ratio 1 : 3. 2
- d) Differentiate
- i. $\frac{4}{\sqrt{x}}$ 1
- ii. $(x^2 - 1)^{10}$ 2
- iii. $\frac{x}{x+1}$ 2

Question 2 START A NEW PAGE

(10 marks)

- a) If $\sin \alpha = \frac{3}{4}$, $0 < \alpha < 90^\circ$ and $\sin \beta = \frac{2}{3}$, $90^\circ < \beta < 180^\circ$,
find the exact values of :
- i. $\tan 2\alpha$ 2
- ii. $\sin(\alpha - \beta)$ 2
- b) Find $\frac{dy}{dx}$ if $y = 2x(x + 4)^8$ 2
- c) Solve the equation $\sin 2x = \tan x$ for $0 \leq x \leq \pi$ 4

Question 3 START A NEW PAGE

(10 MARKS)

a) If α and β are the roots of $3x^2 - 4x + 8 = 0$, find the value of :

i. $\alpha\beta$ 1

ii. $(\alpha - \beta)^2$ 2

b)

i. Show that $\frac{1 + \cos 2A}{\sin 2A} = \cot A$ 2

ii. Hence, find the exact value of $\cot 15^\circ$ 1

c) The curve $y = ax^3 + bx$ cuts the x axis at $x = 1$ and the gradient of the tangent at this point is 4.

i. Find the values of a and b 2

ii. Find the acute angle between the tangent at $x = 1$ and the line $4x - 3y + 2 = 0$ 2

Question 4 START A NEW PAGE

(10 MARKS)

a) For the function $y = \frac{1}{1+x^2}$

i. Find the derivative 2

ii. Determine the equation of the normal at the point where

$y = \frac{1}{2}$ and $x < 0$ 3

iii. Where does this normal cut the x axis? 1

b) Given $\tan \frac{\theta}{2} = t$

i. Express $\sec \theta + \tan \theta$ in terms of t

(express your answer in its simplest form) 2

ii. Hence, or otherwise, solve $\sec \theta + \tan \theta = 2$, correct to nearest

degree for $0^\circ < \theta < 90^\circ$ 2

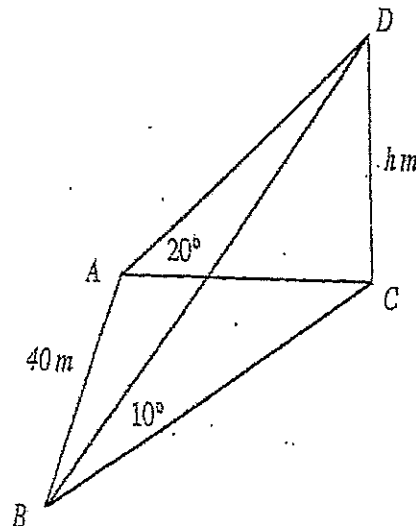
a) Show that $(\sin A - \cos A)^2 = 1 - \sin 2A$ 2

b)

i. Express $\cos \theta - \sqrt{3} \sin \theta$ in the form $R \cos(\theta + \alpha)$ for $R > 0$ 2

ii. Hence, or otherwise, solve $\cos \theta - \sqrt{3} \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$ 2

- c) A vertical flagpole CD of height h metres, stands with its base C on horizontal ground. A is a point on the ground due West of C and B is a point on the ground 40 metres due South of A . From A and B the angles of elevation of the top D of the flagpole are 20° and 10° respectively.



- i. Use triangle ADC to show that $AC = h \tan 70^\circ$ 1
- ii. Write a similar expression for BC 1
- iii. Find the height of the flagpole correct to the nearest metre. 2

- a) For what value (or values) of k , will the quadratic equation
 $(k + 4)x^2 - 3kx - 4(k - 2) = 0$ have two roots which differ by 1.
(hint let the roots be α and $\alpha + 1$) 4
- b)
- i. State the conditions for the quadratic expression $ax^2 + bx + c$
to be negative definite. 1
- ii. Hence, or otherwise, show that the expression
 $(k^2 + k)x^2 - (2k - 6)x + 2$ can never be negative definite 3
- iii. Find the range of values of k for which the expression is positive definite 2

End of examination

Question 1

a) $\tan\left(-\frac{3\pi}{4}\right)$

$= \tan(-135^\circ)$

$= 1 \quad \checkmark$

b) $\cos 105^\circ$

$\cos(60^\circ + 45^\circ) \quad \checkmark$

$= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$

$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$

$= \frac{1 - \sqrt{3}}{2\sqrt{2}} \quad \checkmark$

$= \frac{\sqrt{2} - \sqrt{6}}{4}$

c) $(1, 4)(5, 2)$
 $-1: 3$

pt = $\left(\frac{3 + -5}{2}, \frac{12 + -2}{2}\right)$

$= (-1, 5) \quad \checkmark \checkmark$

d) $\frac{d}{dx} \left(4/\sqrt{x}\right) = -2x^{-3/2}$

$= -2/\sqrt{x^3} \quad \checkmark$

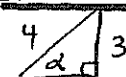
ii. $d/dx (x^2 - 1)^{10}$

$= 20x(x^2 - 1)^9 \quad \checkmark \checkmark$

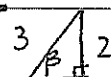
iii. $dy/dx = \frac{(x+1)(1) - (x)(1)}{(x+1)^2}$

$= \frac{1}{(x+1)^2} \quad \checkmark \checkmark$

Question 2

a) 

1st quad



2nd quad

i. $\tan 2\alpha = \frac{2\tan\alpha}{1 - \tan^2\alpha} = \frac{2 \times 3/\sqrt{7}}{1 - 9/7}$

$= \frac{6/\sqrt{7}}{-2/7}$

$= -3\sqrt{7} \quad \checkmark \checkmark$

ii. $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \sin\beta\cos\alpha$

$= \frac{3}{4} \times \frac{\sqrt{5}}{3} - \frac{2}{3} \times \frac{\sqrt{7}}{4}$

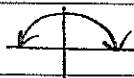
$= \frac{-3\sqrt{5} - 2\sqrt{7}}{12} \quad \checkmark \checkmark$

b) $\frac{dy}{dx} = vu' + uv'$

$= (x+4)^8 \cdot 2 + 2x[8(x+4)^7]$

$= 2(x+4)^7(x+4 + 8x)$

$= 2(9x+4)(x+4)^7 \quad \checkmark \checkmark$

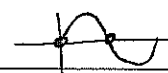
c) $\sin 2x = \tan x$ 

$2\sin x \cos x = \frac{\sin x}{\cos x} \quad \checkmark$

$2\sin x \cos^2 x - \sin x = 0$

$\sin x (2\cos^2 x - 1) = 0 \quad \checkmark$

$\sin x = 0 \quad \cos^2 x = 1/2$

 $\cos x = \pm \frac{1}{\sqrt{2}} \quad \checkmark$

$x = 0, \pi, \pi/4, 3\pi/4$

ie

$x = 0, \pi/4, 3\pi/4, \pi \quad \checkmark$

Question 3

a) $3x^2 - 4x + 8 = 0$

i. $\alpha\beta = c/a = 8/3$ ✓

$$\begin{aligned} \text{ii. } (\alpha - \beta)^2 &= (\alpha + \beta)^2 - 4\alpha\beta \quad \checkmark \\ &= \left(\frac{4}{3}\right)^2 - 4\left(\frac{8}{3}\right) \\ &= -8\frac{8}{9} \quad \checkmark \end{aligned}$$

c) $y = ax^3 + bx - \text{① } (1,0) M_T = 4$

$y' = 3ax^2 + b - \text{②}$

at $(1,0) y' = 4$

$$\therefore 0 = a + b \quad \text{and} \quad \leftarrow \begin{array}{l} \leftarrow - \end{array}$$

$4 = 3a + b$

$2a = 4$

$a = 2$ ✓

$b = -2$ ✓

b) LHS = $\frac{1 + \cos 2A}{\sin 2A}$

$\sin 2A$

$= \frac{1 + (2\cos^2 A - 1)}{2\sin A \cos A} \quad \checkmark$

$2\sin A \cos A$

$= \frac{2\cos^2 A}{2\sin A \cos A}$

$2\sin A \cos A$

$= \frac{\cos A}{\sin A} \quad \checkmark$

$\sin A$

$= \cot A$

$= \text{RHS}$

ii $4x - 3y + 2 = 0$

$3y = 4x + 2$

$y = \frac{4}{3}x + \frac{2}{3}$

$M_1 = 4, M_2 = \frac{4}{3}$

$\tan \alpha = \left| \frac{M_1 - M_2}{1 + M_1 M_2} \right| \quad \checkmark$

$= \left| \frac{4 - \frac{4}{3}}{1 + 4 \times \frac{4}{3}} \right|$

$\alpha = 22^\circ 50'$ ✓

ii. $\cot 15^\circ = \frac{1 + \cos 30^\circ}{\sin 30^\circ}$

$\sin 30^\circ$

$= \frac{1 + \frac{\sqrt{3}}{2}}{\frac{1}{2}}$

$\frac{1}{2}$

$= 2 + \sqrt{3} \quad \checkmark$

Question 4.

$$a) i. y' = -1(x^2+1)^{-2} \cdot 2x = \frac{1+2t+t^2}{1-t^2}$$

$$= \frac{-2x}{(x^2+1)^2} \quad \checkmark$$

$$= \frac{(1+t)^2}{(1-t)(1+t)} \quad t \neq \pm 1$$

$$ii. \text{ When } y = 1/2 \text{ and } x < 0$$

$$\frac{1}{2} = \frac{1}{1+x^2} \text{ then } x = -1$$

$$= \frac{1+t}{1-t} \quad \checkmark$$

$$\text{then } M_T = 1/2 \quad \checkmark$$

$$M_N = -2 \quad \checkmark$$

$$ii. \text{ Now } \sec \theta + \tan \theta = 2$$

$$\text{point } (-1, \frac{1}{2}) \text{ and } M_N = -2$$

$$\frac{1+t}{1-t} = 2$$

$$y - \frac{1}{2} = -2(x+1)$$

$$1+t = 2 - 2t$$

$$2y - 1 = -4(x+1)$$

$$3t = 1$$

$$2y - 1 = -4x - 4 \quad \checkmark$$

$$t = 1/3 \quad \checkmark$$

$$4x + 2y + 3 = 0$$

$$\therefore \tan \theta/2 = 1/3$$

$$\theta/2 = 18^\circ 26'$$

$$iii. x\text{-int } y=0 \therefore x = -3/4$$

$$\theta = 36^\circ 52'$$

$$\div 37^\circ \quad \checkmark$$

$$b) \tan \frac{\theta}{2} = t$$

$$\cos \theta = \frac{1-t^2}{1+t^2}$$

$$\tan \theta = \frac{2t}{1-t^2}$$

$$\therefore \sec \theta + \tan \theta$$

$$= \frac{1+t^2}{1-t^2} + \frac{2t}{1-t^2} \quad \checkmark$$

Question 5

a. $(\sin A - \cos A)^2 = 1 - \sin 2A$

LHS = $(\sin A - \cos A)^2$

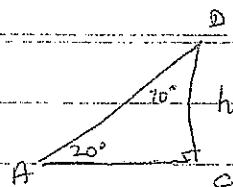
= $\sin^2 A - 2\sin A \cos A + \cos^2 A$

= $1 - 2\sin A \cos A$

= $1 - \sin 2A$

= RHS. ✓

c. i. $\triangle ADC$



$\tan 70^\circ = \frac{AC}{h}$

$AC = h \tan 70^\circ$ ✓

ii. $BC = h \tan 80^\circ$ ✓

b) $\cos \theta - \sqrt{3} \sin \theta$

$R = \sqrt{1^2 + (\sqrt{3})^2}$

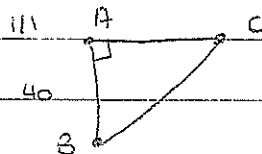
= $\sqrt{4}$

= 2 ✓

$\tan \alpha = \frac{\sqrt{3}}{1}$

$\alpha = \frac{\pi}{3}$ ✓

$\therefore \cos \theta - \sqrt{3} \sin \theta = 2 \cos(\theta + \frac{\pi}{3})$



$BC^2 = 40^2 + AC^2$

$h^2 \tan^2 80^\circ = 1600 + h^2 \tan^2 70^\circ$

$h^2 (\tan^2 80^\circ - \tan^2 70^\circ) = 1600$ ✓

$h^2 = \frac{1600}{\tan^2 80^\circ - \tan^2 70^\circ}$

$\tan^2 80^\circ - \tan^2 70^\circ$

$h^2 = 65.0015 \dots$

ii. Now $2 \cos(\theta + \frac{\pi}{3}) = 1$

$\cos(\theta + \frac{\pi}{3}) = \frac{1}{2}$ ✓

$\theta + \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$

$\theta = 0, \frac{4\pi}{3}, 2\pi$ ✓

$h = 8.0623 \dots$

height = 8 metres ✓

Question 6

a) $(K+4)x^2 - 3Kx - 4(K-2) = 0$

$$\frac{2\alpha + 1}{K+4} = \frac{3K}{K+4}$$

$$\alpha = \frac{K-2}{K+4} \quad \text{--- (1)} \quad \checkmark$$

$$\alpha(\alpha+1) = \frac{-4(K-2)}{K+4} \quad \text{--- (2)} \quad \checkmark$$

Sub (1) into (2)

$$\frac{(K-2)^2}{(K+4)^2} + \frac{(K-2)}{K+4} = \frac{-4K+8}{K+4}$$

$$(K-2)^2 + (K-2)(K+4) = -4(K-2)(K+4) \quad \checkmark$$

$$(K-2)^2 = -5(K-2)(K+4)$$

$$K=2 \text{ is a soln}$$

also $K-2 = -5(K+4)$

$$K-2 = -5K-20$$

$$6K = -18$$

$$K = -3$$

$\therefore K=2 \text{ or } K=-3 \quad \checkmark$

a^2 can't be neg definite

as $a < 0, \Delta < 0$ have no common values of k .

iii. Now $a > 0, \Delta < 0$

i.e. $K^2 + K > 0$

$$K < -1, K > 0$$

with $\Delta < 0$

$$\begin{array}{c} \leftarrow \text{---} 0 \text{---} \rightarrow \\ -9 \quad -1 \quad 0 \quad 1 \end{array}$$

$\checkmark \checkmark$

$\therefore K < -9, K > 1$ is solution for positive definite

b) i Negative definite

$$a < 0, \Delta < 0 \quad \checkmark$$

ii. $\Delta = b^2 - 4ac$

$$(2K-6)^2 - 4(K^2+K)(2) < 0 \quad \checkmark$$

$$4K^2 - 24K + 36 - 8K^2 - 8K < 0$$

$$-4K^2 - 32K + 36 < 0$$

$$K^2 + 8K - 9 \neq 0$$

$$(K+9)(K-1) \neq 0$$

$$K < -9, K > 1 \quad \checkmark$$

but $K^2 + K < 0$

$$K(K+1) < 0$$

$$-1 < K < 0$$

$\checkmark$ with conclusion